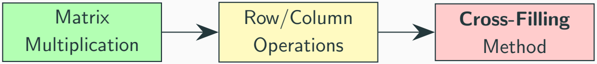
Cross-Filling Method

Rank-One Decomposition and Sum $\leftrightarrow$ Product Equivalence

Introduction



Key Questions:

- What is a **rank-one matrix**?
- How does **cross-filling** decompose any matrix into rank-one pieces?
- How to convert between **sum form** and **product form**?
- What is the **rank** of a matrix?

Review: Sum-of-Rank-One View

Recall from Lecture 1, matrix multiplication has a **sum-of-rank-one** view:

$$C = \begin{pmatrix} | & | & & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & & | \end{pmatrix} \begin{pmatrix} - & b_1^T & - \\ - & b_2^T & - \\ \vdots & \vdots & \vdots \\ - & b_n^T & - \end{pmatrix} = \sum_{k=1}^{n} a_k \, b_k^T$$

Each $a_k \, b_k^T$ is a **rank-one matrix** (column $\times$ row).

Today's question: Can we **reverse** this?

Given any matrix C, decompose it into rank-one pieces?

Motivating Question

Coffee shop revisited:

We have a direct recipe table:

$C =$

	🍱	🍲
🍃	2	2
🍋	1	1
🫘	0	4
🐄	2	1

Question: Can we discover **hidden intermediate products** from just this table?

The algorithm for doing this is called **cross-filling** ().

Rank-One Matrices

Rank-One Matrix

Definition

A matrix R is **rank-one** if $R = \mathbf{u}\,\mathbf{v}^T$ for some nonzero column vector $\mathbf{u}$ and nonzero row vector $\mathbf{v}^T$.

Example:

$$R = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} 1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 4 \\ 1 & 2 \\ 3 & 6 \end{pmatrix}$$

All columns are multiples of $\mathbf{u} = (2, 1, 3)^T$.

All rows are multiples of $\mathbf{v}^T = (1, 2)$.

Recognizing Rank-One Structure

Is this matrix rank-one?

$$\begin{pmatrix} 6 & 9 & 3 \\ 4 & 6 & 2 \\ 10 & 15 & 5 \end{pmatrix}$$

Check rows:

- Row 1: $(6, 9, 3) = 3 \cdot (2, 3, 1)$
- Row 2: $(4, 6, 2) = 2 \cdot (2, 3, 1)$
- Row 3: $(10, 15, 5) = 5 \cdot (2, 3, 1)$

Yes! $R = \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix} \begin{pmatrix} 2 & 3 & 1 \end{pmatrix}.$

Cross-Product Equality

In a rank-one matrix $R = \mathbf{u}\,\mathbf{v}^T$, every entry is $r_{ij} = u_i v_j$.

For any four entries forming a rectangle:

$$\boxed{r_{ij} \cdot r_{k\ell} = r_{i\ell} \cdot r_{kj}}$$

Why? $(u_i v_j)(u_k v_\ell) = (u_i v_\ell)(u_k v_j) = u_i u_k v_j v_\ell$.

Check: In $\begin{pmatrix} 6 & 9 \\ 4 & 6 \end{pmatrix}$: $\quad 6 \times 6 = 36 = 9 \times 4 \quad \checkmark$

Not Every Matrix is Rank-One

$$C = \begin{pmatrix} 2 & 4 \\ 1 & 2 \\ 3 & 7 \end{pmatrix}$$

Check ratios: $\frac{4}{2} = 2, \quad \frac{2}{1} = 2, \quad \frac{7}{3} \neq 2 \quad \leftarrow$ Different!

This is **not rank-one**. But can we write it as a **sum** of rank-one pieces?

$$C = \underbrace{\begin{pmatrix} ? \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} ? \end{pmatrix}}_{R_2}$$

Cross-Filling: One Step

Step 1: Choose a Pivot

$$C = \begin{pmatrix} 2 & 4 \\ 1 & 2 \\ 3 & 7 \end{pmatrix}$$

Choose any nonzero entry as the **pivot**. We pick $c_{11} = 2$:

$$C = \begin{pmatrix} \mathbf{2} & 4 \\ 1 & 2 \\ 3 & 7 \end{pmatrix}$$

Step 2: Extract the Cross

The **cross** through the pivot: its entire **row** and **column**.

$$C = \begin{pmatrix} \mathbf{2} & 4 \\ 1 & 2 \\ 3 & 7 \end{pmatrix}$$

- Cross row: $\begin{pmatrix} \mathbf{2} & 4 \end{pmatrix}$
- Cross column: $\begin{pmatrix} \mathbf{2} \\ 1 \\ 3 \end{pmatrix}$

Step 3: Fill Using the Cross

Build a rank-one matrix using the **multiplication table** principle:

$$R_{k\ell} = \frac{(\text{column})_k \times (\text{row})_\ell}{\text{pivot}}$$

$$R_{11} = \frac{2 \times 2}{\mathbf{2}} = 2 \qquad R_{12} = \frac{2 \times 4}{\mathbf{2}} = 4$$

$$R_{21} = \frac{1 \times 2}{\mathbf{2}} = 1 \qquad R_{22} = \frac{1 \times 4}{\mathbf{2}} = 2$$

$$R_{31} = \frac{3 \times 2}{\mathbf{2}} = 3 \qquad R_{32} = \frac{3 \times 4}{\mathbf{2}} = 6$$

The Rank-One Piece R_1

$$R_1 = \begin{pmatrix} \mathbf{2} & 4 \\ 1 & 2 \\ 3 & 6 \end{pmatrix} = \frac{1}{\mathbf{2}} \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \begin{pmatrix} 2 & 4 \end{pmatrix}$$

This is $\mathbf{u}_1 \mathbf{v}_1^T$ — a rank-one matrix built entirely from the cross!

Step 4: Compute the Remainder

$$C - R_1 = \begin{pmatrix} 2 & 4 \\ 1 & 2 \\ 3 & 7 \end{pmatrix} - \begin{pmatrix} 2 & 4 \\ 1 & 2 \\ 3 & 6 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Key: The **entire cross region** (row 1 and column 1) is now **zero**!

This is why it is called "cross-filling" — the cross is **filled in** and eliminated.

Step 5: The Remainder is Simpler

The remainder $C_2 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}$ is already rank-one:

$$R_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \end{pmatrix}$$

Final decomposition:

$$\underbrace{\begin{pmatrix} 2 & 4 \\ 1 & 2 \\ 3 & 7 \end{pmatrix}}_{C} = \underbrace{\begin{pmatrix} \mathbf{2} & 4 \\ 1 & 2 \\ 3 & 6 \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & \mathbf{1} \end{pmatrix}}_{R_2}$$

Why "Cross-Filling"?

$$\underset{\text{Original}}{\begin{pmatrix} 2 & 4 \\ 1 & 2 \\ 3 & 7 \end{pmatrix}} \xrightarrow{\text{pivot}} \underset{\text{Extract cross}}{\begin{pmatrix} \mathbf{2} & 4 \\ 1 & \cdot \\ 3 & \cdot \end{pmatrix}} \xrightarrow{\text{fill}} \underset{\text{Fill } R_1}{\begin{pmatrix} \mathbf{2} & 4 \\ 1 & 2 \\ 3 & 6 \end{pmatrix}} \xrightarrow{\text{subtract}} \underset{\text{Remainder}}{\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}}$$

1. Choose pivot (center of cross)
2. Extract cross (row + column through pivot)
3. **Fill** all entries via multiplication table
4. Subtract $\rightarrow$ cross region becomes zero

One-Step Cross-Filling: Formula

Proposition

Let A be a matrix with $a_{ij} \neq 0$. The rank-one piece from cross-filling at pivot position (i, j):

$$R = \frac{1}{a_{ij}} \cdot (\text{column } j \text{ of } A) \cdot (\text{row } i \text{ of } A)$$

After subtracting R from A:

- Row i of $A - R$ is all zeros
- Column j of $A - R$ is all zeros

The cross through (i, j) is **eliminated**.

Cross-Filling Algorithm

A 3×3 Example

Decompose the following matrix by cross-filling:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix}$$

We will peel off rank-one pieces iteratively until nothing remains.

Iteration 1: Pivot $a_{11} = 1$

Highlight the cross through a_{11}:

$$A = \begin{pmatrix} \mathbf{1} & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix}$$

Fill using the cross:

$$R_1 = \frac{1}{\mathbf{1}} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \mathbf{1} & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$

Iteration 1: Remainder A_1

$$A_1 = A - R_1 = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix} - \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 7 \end{pmatrix}$$

Row 1 and column 1 are now zero. ✓

Iteration 2: Pivot $(A_1)_{22} = 2$

$$A_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{2} & 2 \\ 0 & 2 & 7 \end{pmatrix}$$

Fill using the new cross:

$$R_2 = \frac{1}{\mathbf{2}} \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} \begin{pmatrix} 0 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{2} & 2 \\ 0 & 2 & 2 \end{pmatrix}$$

Iteration 2: Remainder A_2

$$A_2 = A_1 - R_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 7 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 5 \end{pmatrix}$$

Rows 1–2 and columns 1–2 are now zero. ✓

Iteration 3: Final Piece

$$A_2 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \mathbf{5} \end{pmatrix}$$

This is already rank-one:

$$R_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \mathbf{5} \end{pmatrix}$$

Remainder: $A_3 = A_2 - R_3 = O$. **Done!**

Full Decomposition

$$A = \underbrace{\begin{pmatrix} \mathbf{1} & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}}_{R_1} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{2} & 2 \\ 0 & 2 & 2 \end{pmatrix}}_{R_2} + \underbrace{\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \mathbf{5} \end{pmatrix}}_{R_3}$$

Three rank-one pieces:

- R_1: pivot **1** at position $(1, 1)$
- R_2: pivot **2** at position $(2, 2)$
- R_3: pivot **5** at position $(3, 3)$

The Cross-Filling Algorithm

Proposition (Cross-Filling Algorithm)

Input: Matrix A ($m \times n$).

Output: $A = R_1 + R_2 + \cdots + R_r$ (sum of rank-one matrices).

Procedure:

1. Find any nonzero entry $a_{ij} \neq 0$ (the **pivot**).
2. Form rank-one piece:

$$R = \frac{1}{a_{ij}} \cdot (\text{column } j) \cdot (\text{row } i)$$

3. Compute remainder $A' = A - R$.
4. Repeat on A' until $A' = O$.

Each step eliminates one cross (one row and one column become zero).

Rank of a Matrix

Definition

The **rank** of a matrix A is the number of rank-one pieces in its cross-filling decomposition:

$$A = R_1 + R_2 + \cdots + R_r \quad \Longrightarrow \quad \text{rank}(A) = r$$

Examples:

- Zero matrix O: rank 0
- Rank-one matrix $\mathbf{u}\mathbf{v}^T$: rank 1
- Identity I_n: rank n
- Our A above: rank 3

Warning: Different pivot choices give different decompositions. Do they always give the same count? **Yes** — but we prove this in Lecture 4.

Sum $\leftrightarrow$ Product Equivalence

Outer Product Recap

Each rank-one piece is an **outer product** (column $\times$ row):

$$R_k = \mathbf{u}_k \, \mathbf{v}_k^T$$

	Inner product	Outer product
Form	$\mathbf{u}^T \mathbf{v}$ (row $\times$ col)	$\mathbf{u}\,\mathbf{v}^T$ (col $\times$ row)
Result	scalar	matrix
Size	$(1\times n)(n\times 1) = 1\times 1$	$(m\times 1)(1\times n) = m\times n$

Cross-filling produces **outer products** — each is a rank-one matrix.

Collecting the Pieces

From our 3×3 example:

$$R_1 = \underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{\mathbf{u}_1} \underbrace{\begin{pmatrix} 1 & 1 & 1 \end{pmatrix}}_{\mathbf{v}_1^T}$$

$$R_2 = \underbrace{\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}}_{\mathbf{u}_2} \underbrace{\begin{pmatrix} 0 & 2 & 2 \end{pmatrix}}_{\mathbf{v}_2^T}$$

$$R_3 = \underbrace{\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}_{\mathbf{u}_3} \underbrace{\begin{pmatrix} 0 & 0 & 5 \end{pmatrix}}_{\mathbf{v}_3^T}$$

Product Form: $A = UV$

Collect columns $\mathbf{u}_k$ into U, rows $\mathbf{v}_k^T$ into V:

$$U = \begin{pmatrix} | & | & | \\ \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \\ | & | & | \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}$$

$$V = \begin{pmatrix} - & \mathbf{v}_1^T & - \\ - & \mathbf{v}_2^T & - \\ - & \mathbf{v}_3^T & - \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 5 \end{pmatrix}$$

By sum-of-rank-one: $UV = \sum_{k=1}^{3} \mathbf{u}_k \, \mathbf{v}_k^T = R_1 + R_2 + R_3 = A$.

Verification

$$UV = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix} = A \quad \checkmark$$

Notice: U is **lower triangular**, V is **upper triangular**!

This happened because we chose **diagonal pivots** $(1,1) \to (2,2) \to (3,3)$.

Diagonal pivots $\implies$ the product form is an **LU decomposition**.

Sum $\leftrightarrow$ Product Equivalence

Proposition

For any matrix A:

$$A = \sum_{k=1}^{r} \mathbf{u}_k \, \mathbf{v}_k^T \quad \Longleftrightarrow \quad A = U \cdot V$$

where U has columns $\mathbf{u}_k$ and V has rows $\mathbf{v}_k^T$.

Two directions:

- **Lecture 1**: Given product $A = UV$, read off sum $A = \sum \mathbf{u}_k \, \mathbf{v}_k^T$
- **Today**: Given sum from cross-filling, assemble product $A = UV$

The Inner Dimension

When $A = UV$ with A being $m \times n$:

$$A_{m\times n} = U_{m\times \boxed{r}} \cdot V_{\boxed{r}\times n}$$

The shared dimension r is the **inner dimension**.

- Outer dimensions m, n: visible (size of A)
- Inner dimension r: **hidden** $=$ number of independent patterns

Key: $\text{rank}(A) = r =$ inner dimension of the factorization.

Algebraic Expression for Cross-Filling

Cross-Filling as a Matrix Formula

Let e_i be the i-th column of I_n, and e_j^T the j-th row of I_m.

Then:

- Ae_i = column i of A
- $e_j^T A$ = row j of A
- $e_j^T A e_i$ = entry (j, i) of A = the pivot

The cross-filling formula at pivot position (row j, column i):

$$\boxed{R = Ae_i \, (e_j^T A e_i)^{-1} \, e_j^T A}$$

This is: $\dfrac{1}{\text{pivot}} \times (\text{column}) \times (\text{row})$.

Example Verification

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 3 & 3 \\ 1 & 3 & 8 \end{pmatrix}, \quad \text{cross-fill at row 1, column 1:}$$

$$Ae_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad e_1^T A = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}, \quad e_1^T A e_1 = 1$$

$$R_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot 1 \cdot \begin{pmatrix} 1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \quad \checkmark$$

Application: Projection Matrices

Definition

A square matrix P is a **projection** if $P^2 = P$.

Proposition

If P is a projection and R is obtained by cross-filling from P, then R is also a projection.

Proof: Let $R = Pe_i(e_j^T Pe_i)^{-1}e_j^T P$. Then

$$R^2 = Pe_i(e_j^T Pe_i)^{-1} \underbrace{e_j^T P \cdot Pe_i}_{= e_j^T Pe_i}(e_j^T Pe_i)^{-1}e_j^T P$$

$$= Pe_i(e_j^T Pe_i)^{-1}e_j^T P = R \quad \checkmark$$

(Used $P^2 = P$ in the underbraced step.)

Projection Decomposition

If P is a projection and R comes from cross-filling P:

$$P = \underbrace{R}_{\text{projection}} + \underbrace{(P - R)}_{\text{also projection!}}$$

Check: $(P - R)^2 = P^2 + R^2 - PR - RP = P + R - R - R = P - R \quad \checkmark$

(Here $PR = QP = R$ since $P^2 = P$ implies columns and rows of P are fixed by P.)

Cross-filling **decomposes projections into smaller projections**.

This will be **extremely important** for spectral decomposition.

Preview: LU Decomposition

Diagonal Pivots $\Rightarrow$ LU

In our example, diagonal pivots produced:

$$A = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}}_{\text{lower triangular } L} \underbrace{\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 5 \end{pmatrix}}_{\text{upper triangular } U}$$

Key: The pivots $\mathbf{1}, \mathbf{2}, \mathbf{5}$ appear on the diagonal of U.

Diagonal cross-filling $\Longrightarrow$ **LU decomposition**.

LDU Decomposition

We can further separate the pivots into a diagonal matrix:

$$A = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{pmatrix}}_{L} \underbrace{\begin{pmatrix} \mathbf{1} & 0 & 0 \\ 0 & \mathbf{2} & 0 \\ 0 & 0 & \mathbf{5} \end{pmatrix}}_{D} \underbrace{\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}}_{U}$$

- L: lower triangular, 1's on diagonal (from normalized cross columns)
- D: diagonal matrix of **pivots** (the cross centers)
- U: upper triangular, 1's on diagonal (from normalized cross rows)

Each rank-one piece: $R_k = LD_kU$ where D_k keeps only the k-th pivot.

Not All Matrices Are LDU-Decomposable

Diagonal cross-filling requires **nonzero diagonal pivots**. But this can fail:

$$A = \begin{pmatrix} 0 & 3 \\ 1 & 2 \end{pmatrix}$$

Pivot $a_{11} = 0$ — cannot start!

Even later pivots can fail:

$$\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 1 & 3 & 5 \end{pmatrix} \xrightarrow{\text{after } R_1} \begin{pmatrix} 0 & 0 & 0 \\ 0 & \mathbf{0} & 1 \\ 0 & 1 & 2 \end{pmatrix}$$

Second pivot is zero!

Solution: Allow non-diagonal pivots, or permute rows first $\rightarrow$ **PLU decomposition** (future lecture).

Exercises

Exercise 1

Exercise 1.1: Cross-fill the following matrix. How many rank-one pieces are needed?

$$A = \begin{pmatrix} 2 & 6 & 4 \\ 1 & 3 & 2 \end{pmatrix}$$

Exercise 1.2: Cross-fill using diagonal pivots, and write the result as $A = LU$.

$$B = \begin{pmatrix} 2 & 4 & 2 \\ 1 & 3 & 2 \\ 4 & 10 & 8 \end{pmatrix}$$

Exercise 2

Exercise 1.3: Let A be an $m \times n$ matrix with $\mathrm{rank}(A) = r$.

(a) Explain why $r \leq \min(m, n)$.

(b) Show that $A = UV$ where U is $m \times r$ and V is $r \times n$.

Exercise 3

Exercise 1.4: Let P be a 3×3 projection matrix ($P^2 = P$):

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(a) Cross-fill at pivot $p_{11} = \frac{1}{2}$. Find the rank-one piece R.

(b) Verify $R^2 = R$.

(c) Verify $(P - R)^2 = P - R$.

Summary

Today we learned:

1. **Rank-one matrix**: $R = \mathbf{u}\,\mathbf{v}^T$ (one pattern, repeated)
2. **Cross-filling algorithm**: Decompose any matrix into rank-one pieces

$$A = R_1 + R_2 + \cdots + R_r$$

3. **Sum $\leftrightarrow$ Product**:

$$\sum_{k=1}^{r} \mathbf{u}_k\,\mathbf{v}_k^T = A = UV$$

4. **Rank** = number of pieces = inner dimension of UV
5. **Algebraic formula**: $R = Ae_i(e_j^T A e_i)^{-1} e_j^T A$
6. **Diagonal pivots** $\Rightarrow$ LU decomposition